

Simplified Gaussian Model for the Open-Source Coverage Tool

John Albertson

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1 Model objective

The open-source coverage tool uses a simple Gaussian plume model to estimate whether a methane source is detectable by a fixed sensor for a given wind speed, wind direction, source height, sensor height, and reference release rate.

This version intentionally does not include downwash, wake enhancement, or obstacle-resolved flow.

2 Gaussian plume model with ground reflection

For a source located upwind of a sensor, the modeled methane concentration enhancement at downwind distance x is

$$C(x) = \frac{Q_{\text{ppm}}}{2\pi U \sigma_y(x) \sigma_z(x)} \left[\exp\left(-\frac{(Z_s - H)^2}{2\sigma_z^2(x)}\right) + \exp\left(-\frac{(Z_s + H)^2}{2\sigma_z^2(x)}\right) \right].$$

Symbol	Definition
$C(x)$	modeled methane concentration enhancement [ppm]
x	downwind source-to-sensor distance [m]
U	wind speed [m/s]
H	source height [m]
Z_s	sensor height [m]
$\sigma_y(x)$	lateral plume spread [m]
$\sigma_z(x)$	vertical plume spread [m]
Q_{ref}	methane reference release rate [kg/hr]
Q_{ppm}	release-rate conversion used in the ppm plume equation

The second exponential term is the standard ground-reflection term.

The release-rate conversion used in the current implementation is

$$Q_{\text{ppm}} = \frac{Q_{\text{ref}}}{3.6} \frac{V_m}{MW_{\text{CH}_4}} 1000,$$

with

$$V_m = 24.45 \text{ L/mol}, \quad MW_{\text{CH}_4} = 16.04 \text{ g/mol}.$$

Detection occurs when

$$C(x) \geq C_{\text{MDL}},$$

where C_{MDL} is the sensor minimum detection level [ppm]. The downwind locations where

$$C(x) = C_{\text{MDL}}$$

define the detection interval,

$$x_{\min} \leq x \leq x_{\max}.$$

3 Smooth Pasquill-Gifford plume spreads

The plume spreads are computed using the smooth open-country Pasquill-Gifford relationships reported by Hanna (1984).

$$\sigma_y(x) = a_y x (1 + b_y x)^{-1/2},$$

$$\sigma_z(x) = a_z x (1 + b_z x)^{-p}.$$

Here x , σ_y , and σ_z are in meters.

PG class	Description	a_y	b_y	a_z	b_z	p
A	Very unstable	0.22	0.0001	0.200	0.0000	0
B	Unstable	0.16	0.0001	0.120	0.0000	0
C	Slightly unstable	0.11	0.0001	0.080	0.0002	1/2
D	Neutral	0.08	0.0001	0.060	0.0015	1/2
E	Slightly stable	0.06	0.0001	0.030	0.0003	1
F	Stable	0.04	0.0001	0.016	0.0003	1

4 Recommended default for the open-source tool

Although the table above gives all six PG classes, the recommended default for the open-source coverage tool is to use the neutral PG-D case:

$$\sigma_y(x) = 0.08x(1 + 0.0001x)^{-1/2},$$

$$\sigma_z(x) = 0.06x(1 + 0.0015x)^{-1/2}.$$

This choice is based on the stability analysis performed for representative eddy-covariance sites. Those results showed that roughly 60% of conditions were PG-D neutral, and roughly 85% were near-neutral when PG-C, PG-D, and PG-E were grouped together.

We also compared the detection envelope from a full six-class PG calculation, weighted by annual stability fractions, with the neutral-only PG-D envelope. The main practical detection ranges were similar. The differences were mainly low-probability tails associated with less frequent strongly stable or unstable conditions.

Therefore, for the first open-source siting tool, use PG-D only. This keeps the model simple, transparent, and suitable for annual screening-level coverage calculations.

References

Hanna, S. R. (1984). Applications in Air Pollution Modeling. In *Atmospheric Turbulence and Air Pollution Modelling*, pp. 275–310. Springer Netherlands.